COLIN NGUYEN

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AI Engineer building production LLM agents and the evaluation infrastructure that keeps them reliable in regulated fintech: agent design and orchestration, rigorous LLM evaluation, and coding-agent workflows, owned end to end.

EXPERIENCE

Athena Money *Nov 2025 – Present*
Founding AI Engineer *San Francisco Bay Area*

- Designed and built the team's LLM evaluation platform for multi-turn agents in regulated mortgage underwriting: scenario graders, rubric-based LLM-as-judge scoring, per-turn quality scoring and outcome checkpoints, driven by an LLM user simulator.
- Built field-level consensus evaluation (an agent run across repeated trials with per-field agreement and confidence-score calibration) plus statistical regression gating in tiered CI; eval driver on durable Temporal workflows for reproducible reruns.
- Shipped production underwriting and follow-up agents on Google ADK and Temporal: every action validated in the agent's turn, failing closed, with a supervising layer and a borrower override path; migrated the legacy workflow onto them behind flags, then retired it. Added scheduled coding agents that audit the product and open pull requests behind an enforced merge boundary.
- Built the joint-application lifecycle end to end across backend, web and both mobile apps: inviting or standing in for a co-borrower, joining at rate lock, per-applicant disclosures, and routing identity mismatches to review.
- Took the Android banking app (Kotlin, Jetpack Compose) from a partial shell to parity with iOS; built snapshot-test harnesses on both platforms as blocking merge checks and parallelised the Android CI critical path.

Zeta Global *Jun 2025 – Aug 2025*
Data Science & Generative AI Intern *San Francisco*

- Engineered a LangGraph supervisor agent that routes tasks to sub-agents, cutting manual workflow setup time by 70%.
- Built a multi-vendor (OpenAI, Anthropic, Bedrock) orchestration platform on FastAPI with PostgreSQL checkpointing, plus an evaluation framework (accuracy, consistency, latency, cost); deployed via REST APIs on Docker and Kubernetes.

NationBuilder *Jan 2025 – Nov 2025*
AI Research Intern (UCSB capstone project through Jun 2025) *Remote*

- RAG pipeline (all-MiniLM-L6-v2 embeddings, FAISS) ranking document relevance at 93% relevance-classification accuracy.
- Capstone: scraped election data and built an agentic PDF-extraction system (GPT-4o) with strict JSON schemas.

RESEARCH

Risk-Controlled Decision Gates — with Prof. Eric Suess, CSU East Bay *Oct 2026 – Present*

- Pre-registered study of whether a Learn-then-Test abstention guarantee survives option-count shift: it fails from 5 to 150 options on a public decision model and, with its mechanism, on SmolLM2 models tuned to match it; four calibration regimes compared.

Efficient LLM Fine-Tuning — with Prof. Eric Suess, CSU East Bay *Sep 2026 – Oct 2026*

- Five stacked memory-saving techniques raised SmolLM2's trainable step from 8,192 to 131,072 tokens in 71% of baseline memory; a hand-written block-wise 8-bit AdamW matched torchao's to the byte and fp32 across six seeds (pre-registered TOST).

Evaluation of Mobile-Health Receptivity Models — with R. Benzo (OSU) and P. Washington (UCSF) *Jun 2026 – Present*

- Showed that the standard pooled metric for receptivity models is dominated by between-person comparisons across three cohorts, so it mostly rewards knowing who tends to respond; pre-registered personalisation study; manuscript in preparation.

SELECTED PROJECTS

Pokémon TCG AI Battle Challenge — Kaggle *Aug 2026 – Sep 2026*

- Agent for a hidden-information card game on a live ranked ladder: scripted expert policy, determinized whole-turn beam search, and a value network trained by self-play.

LLM 20 Questions — Kaggle *May 2024 – Aug 2024*

- Two-model agent on 4-bit quantized LLMs (Transformers, BitsAndBytes): one asks and guesses, one answers.

SKILLS

Languages: Python, Go, SQL, Java, JavaScript, R, C
LLM / AI: multi-agent orchestration (LangGraph, Google ADK, Temporal), RAG, vector search (FAISS), embeddings, LLM evaluation / LLM-as-judge, MCP (Model Context Protocol), prompt engineering, LangSmith, OpenAI / Anthropic / Bedrock APIs
ML: PyTorch, fine-tuning / PEFT (LoRA, QLoRA), vLLM, scikit-learn, TensorFlow, Node2Vec, feature engineering, calibration
Infra / MLOps: AWS (Bedrock), Google Cloud, Docker, Kubernetes, FastAPI, PostgreSQL, CI/CD, observability, Spark, Databricks

EDUCATION

University of California, Santa Barbara *Sep 2022 – Jun 2025*
B.S. in Statistics & Data Science, Minor in Mathematics *Santa Barbara, CA*

PATENTS

- "Adaptive Neural Content Generation", U.S. Provisional App. No. 63/943,780, filed Dec 2025; first-named inventor.